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United States Patent Application Publication

20250277316

Kind Code

A1

Publication Date

September 04, 2025

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Pressurised Electrolyser

Abstract

The present disclosure relates to an electrolyzer for generating hydrogen, the electrolyzer comprising: a housing comprising an electrolyte chamber; two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, wherein the at least one permeable electrode has a first surface facing the electrolyte chamber and a second surface facing a first gas collection chamber; an electrolyte supply circuit for supplying electrolyte water to the electrolyte chamber; and a control unit and/or mechanical control for controlling a pressure drop across the at least one permeable electrode, between the electrolyte chamber and the first gas collection chamber.

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Family ID: 78718490

Appl. No.: 18/702530

Filed (or PCT Filed): October 11, 2022

PCT No.: PCT/EP2022/078247

Foreign Application Priority Data

GB

2115054.5

Oct. 20, 2021

Publication Classification

Int. Cl.: C25B9/05 (20210101); C25B1/04 (20210101); C25B11/031 (20210101); C25B11/046 (20210101); C25B15/023 (20210101); C25B15/08 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. national phase application filed under 35 U.S.C. § 371 of International Application Number PCT/EP2022/078247, filed Oct. 11, 2022, designating the United States, which claims priority from United Kingdom Patent Application Number, GB 2115054.5, filed Oct. 20, 2021, both of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

[0002] The present disclosure relates to an electrolyzer, particularly but not exclusively, for generating hydrogen. Another aspect of the present disclosure relates to a method of controlling an electrolyzer.

[0003] The process of using electricity to decompose water into oxygen and hydrogen gas is known as electrolysis of water. Hydrogen gas produced in this way can be used in various applications and has become widely known as an energy dense option for fuelling vehicles. In other applications, electrolysis of water may be used as a decentralized storage solution storing electrical energy as chemical energy, particularly electrical energy obtained via renewable power. In recent years, therefore, demand for hydrogen, inter alia, as a fuel for so called hydrogen fuel cells has increased rapidly.

[0004] Electrolyzers can be grouped into proton exchange membrane (PEM) electrolyzer, alkaline electrolyzers and solid oxide electrolyzers. These different types of electrolyzers function in slightly different ways depending on the electrolyte material involved. Yet, some of the most prominent drawbacks of most electrolyzers include overall inefficiencies and/or failure to supply hydrogen gas at pressures required for further use.

[0005] It is an aim of the present disclosure to solve or at least ameliorate one or more problems of the prior art.

[0006] Aspects and embodiments of the present disclosure provide an electrolyzer for generating hydrogen and a method of controlling an electrolyzer for generating hydrogen from water as claimed in the appended claims.

[0007] According to a first aspect of the present disclosure, there is provided an electrolyzer for generating hydrogen, the electrolyzer comprising: [0008] a housing comprising an electrolyte chamber; [0009] two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, wherein the at least one permeable electrode has a first surface facing the electrolyte chamber and a second surface facing a first gas collection chamber; [0010] an electrolyte supply circuit for supplying electrolyte water to the electrolyte chamber; and [0011] a control unit for controlling a pressure drop across the at least one permeable electrode, between the electrolyte chamber and the first gas collection chamber.

[0012] According to one embodiment, the control unit is configured to: [0013] receive gas-pressure-data representative of a gas pressure within the first collection chamber; [0014] determine, on the basis of the gas-pressure-data, a desired electrolyte water pressure within the electrolyte chamber, [0015] control the electrolyte supply circuit to set the pressure within the electrolyte chamber to the desired electrolyte pressure.

[0016] In another embodiment, the control unit is configured to set the desired electrolyte pressure to be higher than a gas pressure within the first gas collection chamber, wherein a difference between the electrolyte pressure and the gas pressure is above a selectable first pressure-threshold.

[0017] In another embodiment, the pressure threshold is at least 5 bar.

[0018] In another embodiment, the control unit is configured to apply a current across the electrodes if the pressure difference between the first gas collection chamber and the electrolyte chamber is above the selectable pressure-threshold.

[0019] In another embodiment, the first gas collection chamber comprises a pressure control valve for regulating a gas pressure within the first gas collection chamber.

[0020] In another embodiment, the electrolyte supply circuit comprises a hydraulic pump for supply of electrolyte water to the electrolyte chamber, and wherein the control unit is configured to: [0021] receive electrolyte-pressure-data representative of a pressure of electrolyte water within the electrolyte chamber; and [0022] control, on the basis of the electrolyte-pressure-data, an operation of the pump.

[0023] In another embodiment, the at least one permeable electrode has a porosity of 0.3 μm or less, preferably 0.2 μm or less.

[0024] In another embodiment, both electrodes are permeable to gases produced by the decomposition of electrolyte water, and wherein the electrodes have different porosities.

[0025] In another embodiment, the at least one permeable electrode is made of steel, preferably sintered steel.

[0026] In another embodiment, the two electrodes are made from different materials.

[0027] In another embodiment, at least one of the electrodes, preferably the cathode, is magnetic.

[0028] In another embodiment, the electrolyte supply circuit comprises a pressure storage reservoir selectively connectable to the electrolyte chamber.

[0029] In another embodiment, the pressure storage reservoir is connected to the electrolyte chamber via a normally-open outlet valve, which is closed during normal operation of the electrolyzer.

[0030] In another embodiment, the control unit is configured to: [0031] receive a forward-flushing command; [0032] interrupt any current applied across the electrodes and increase a pressure in the electrolyte chamber to be more than 5 bar above a gas pressure within the first gas chamber in response to the forward-flushing command.

[0033] In another embodiment, the at least one permeable electrode comprises a thickness between its first and second surface of 3 mm to 100 mm.

[0034] In another embodiment, the electrolyte chamber comprises an electrolyte gap located between the two electrodes, wherein the gap preferably has a width of 0.1 mm to 10 mm.

[0035] In another embodiment, the first gas collection chamber comprises a first drain port for draining electrolyte water that has passed through the at least one permeable electrode back into the electrolyte supply circuit.

[0036] In another embodiment, the control unit is configured to: [0037] receive a backward-flushing-command; [0038] interrupt any current applied across the electrodes and supply electrolyte water to the first gas collection chamber via the first drain port to reverse flush the at least one permeable electrode from the first gas collection chamber towards the electrolyte chamber.

[0039] According to another embodiment aspect of the present disclosure, there is provided a method of controlling an electrolyzer for generating hydrogen from water, the electrolyzer comprising two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, the method comprising: [0040] supplying electrolyte water to the at least one electrode; [0041] controlling a pressure drop between opposite sides of the at least one electrode.

[0042] According to an embodiment, the method comprises maintaining a pressure drop between opposite sides of the at least one electrode that is higher than a selectable pressure-threshold.

[0043] According to an embodiment, the method comprises applying electricity across the electrodes if the pressure drop across opposite sides of the at least one electrode is higher than the

selectable pressure-threshold.

[0044] In another embodiment, the selectable pressure threshold is 5 bar or more.

[0045] According to a first aspect of the present disclosure, there is provided an electrolyzer for generating hydrogen, the electrolyzer comprising: [0046] a housing comprising an electrolyte chamber; [0047] two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, wherein the at least one permeable electrode has a first surface facing the electrolyte chamber and a second surface facing a first gas collection chamber, wherein the at least one permeable electrode has a porosity of 0.3 μm or less.

[0048] Within the scope of this application it is expressly intended that the various aspects, embodiments, examples and alternatives set out in the preceding paragraphs, and the claims and/or the following description and drawings, and in particular the individual features thereof, may be taken independently or in any combination. That is, all embodiments and all features of any embodiment can be combined in any way and/or combination, unless such features are incompatible. The applicant reserves the right to change any originally filed claim or file any new claim accordingly, including the right to amend any originally filed claim to depend from and/or incorporate any feature of any other claim although not originally claimed in that manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] The aforementioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure will be better understood by reference to the following description of embodiments of the disclosure taken in conjunction with the accompanying drawings, wherein:

[0050] FIG. 1 shows a schematic cross-section of an electrolyzer according to an embodiment of the present disclosure;

[0051] FIG. 2 shows a schematic flow chart of a method according to an embodiment of the present disclosure.

[0052] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.

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DETAILED DESCRIPTION

[0054] FIG. 1 shows an electrolyzer **100** according to an embodiment of the present disclosure. In the example of FIG. 1, there is shown an electrolyzer **100** for generating hydrogen, particularly by electrolysis of water. However, as will be appreciated, the electrolyzer **100** shown in FIG. 1 may also be suitable for decomposition of other substances by means of electrolysis. Generally, in this specification, the term “electrolyte water” may encompass water including any type of electrolyte, such as sulphuric acid, sulphate, potassium hydroxide, sodium hydroxide, etc., or in some embodiments water without electrolytes such as tritiated water.

[0055] The electrolyzer **100** comprises a housing **102**. The housing **102** comprises an electrolyte chamber **104**. In one example, the electrolyte chamber is configured to hold electrolyte water under pressure as will be described in more detail below. The electrolyte water may comprise any electrolyte dissolved in water that aids decomposition of water.

[0056] The housing **102** comprises a first gas collection chamber **106**. The first gas collection chamber **106** is separated from the electrolyte chamber **104** by a first electrode **108**. The housing **102** comprises a second gas collection chamber **110**. The second gas collection chamber **110** is separated from the electrolyte chamber **104** by a second electrode **112**. The electrolyte chamber **104**

is located between the first and second electrodes **108**, **112**. The electrolyte chamber **104** is a membrane-less chamber.

[0057] It will be appreciated that in some embodiments, the electrolyzer may comprise more than two electrodes, e.g. arranged in parallel to each other to form a stack of electrodes. The pair of electrodes **108**, **112** of FIG. **1** are thus exemplary for any number of electrodes used in a stack.

[0058] The housing **102** of FIG. **1** is a multi-part housing. In particular, the housing **102** comprises at least two parts: a first housing part includes the first gas collection chamber **106** and the first electrode **108**; a second housing part includes the second gas collection chamber **110** and the second electrode **112**. The at least two housing parts are connected to each other such that a gap is formed between first surfaces **114**, **118** of the first and second electrodes **108**, **112**. This gap defines the electrolyte chamber **104**, which is thus arranged between the electrodes **108**, **112**. In the example of FIG. **1**, the housing **102** is substantially H-shaped as will be described in more detail below. However as will be appreciated.

[0059] text missing or illegible when filed other designs may include a plurality of plate shaped electrodes that typically are flat and similar in construction to a plate heat-exchanger.

[0060] The first electrode **108** is permeable to gases produced by decomposition of electrolyte water. The first electrode **108** is also generally permeable to electrolyte water depending on a pressure within the electrolyte chamber **104** as will be explained in more detail below. In other words, the first electrode **108** is a so-called “flow-through” electrode in which gases produced by decomposition of electrolyte water (and the electrolyte water) within the electrolyte chamber **104** are able to penetrate the first electrode **108** and thus move from the electrolyte chamber **104** towards the first gas collection chamber **106**. To this end, the first electrode **108** is a permeable electrode including a plurality of pores sized to allow permeation of the respective gas, e.g. hydrogen, through the first electrode **108**.

[0061] The first electrode **108** has a first surface **114** facing the electrolyte chamber and a second, opposite surface **116** facing the first gas collection chamber **106**.

[0062] The first electrode **108** may be made of steel, preferably sintered steel. In some embodiments, the second electrode **112** may also be made of steel, preferably sintered steel. In alternative embodiments, the second electrode **112** may be made of a material different to the first electrode **108**.

[0063] Gases produced by the first electrode **108** may flow into the first gas collection chamber **106**. The first gas collection chamber **106** comprises a first gas outlet port **122** for extraction of the gas within the first gas collection chamber **106**. As will be described in more detail below, the first gas outlet port **122** may include a pressure control valve, such as a pressure relief valve, configured to set a gas pressure within the first gas collection chamber **106**.

[0064] The second electrode **112** of FIG. **1** is also permeable to gases produced by the decomposition of electrolyte water. The second electrode **112** is also generally permeable to electrolyte water depending on a pressure within the electrolyte chamber **104** as will be explained in more detail below. In other words, the second electrode **112** is also a “flow-through” electrode in which gases produced by decomposition of electrolyte water within the electrolyte chamber **104** are able to penetrate the second electrode **112** and thus move from the electrolyte chamber **104** towards the second gas collection chamber **110**.

[0065] The second electrode **112** has a first surface **118** facing the electrolyte chamber **104** and a second, opposite surface **120** facing the second gas collection chamber **110**.

[0066] Gases produced by the second electrode **112** may flow into the second gas collection chamber **110**. The second gas collection chamber **110** comprises a second gas outlet port **124** for extraction of the gas within the second gas collection chamber **110**. As will be described in more detail below, the second gas outlet port **124** may include a pressure control valve, such as a pressure relief valve, configured to determine the pressure within the second gas collection chamber **110**.

[0067] The electrolyzer **100** comprises an electrolyte water supply circuit **130** for supplying the electrolyte chamber **104** with electrolyte water. The electrolyte water supply circuit **130** of FIG. **1** is a closed system. Alternatively, the water supply circuit may be an open system including an electrolyte water reservoir. In one example, the electrolyte water supply circuit **130** comprises a pump **134** arranged upstream of an inlet port **126** of the electrolyte chamber **104**. The pump **134** may be configured to move electrolyte water through the system and in the direction of the electrolyte chamber **104** at a selectable pressure.

[0068] The electrolyte supply circuit **130** comprises an electrolyte water supply line **162** for topping up electrolyte water turned into gas by the electrolysis process. As indicated in FIG. **1**, the electrolyte water supply line **162** may be connected to the pump **134** via a manually or automatically controlled shut off valve. An expansion vessel **132** is a pressure storage device arranged within the electrolyte water supply circuit **130** and configured to provide expansion derived pressure into the system and configured to enable **134** to have a lower cycle time and maintain a desired pressure range within the circuit **130**/protect the circuit **130** from excessive pressures.

[0069] The electrolyte chamber **104** comprises a vent port connected to a vent line **105**. The vent line **105** is configured to be used to drain the back flushed electrolyte and any associated media. Additionally, the line may be instrumental in sampling and even comprise a collection well for removal of electrolyte in the circulation system either with or without pressure. Additionally, the vent may or may not be instrumental in the purging of air from the system prior to current being applied to the electrodes. Finally, the vent pipe may allow electrolyte flow and recycling from **104** via **105** to **162**, the water inlet line or it may be discharged/collected.

[0070] The electrolyte supply circuit **130** may comprise a pressure gauge **134** for monitoring the pressure within the electrolyte chamber **104**. As will be appreciated, the pressure gauge **136** may be arranged anywhere downstream of the pump **134**. In some embodiments, the pressure gauge **136** may be an integral part of the pump **134**. In other embodiments, the pressure gauge may be arranged within the electrolyte chamber **104**.

[0071] In some embodiments, the electrolyte water supply circuit **130** may comprise a pressure storage device, such as an accumulator **138**. The accumulator **138** shown in FIG. **1** is arranged downstream of the pump **134**. The accumulator **138** may be connected to the pump **134** via a check valve, which, during normal operation of the electrolyzer **100**, allows fluid to be pumped into the accumulator **138**. As will be appreciated, the check valve will stop fluid from leaving the accumulator **138**. The accumulator **138** may also comprise a separate outlet valve for selectively connecting an outlet of the accumulator **138** with the electrolyte chamber **104**. In some embodiments, the accumulator **138** may thus comprise a normally-open outlet valve, which is closed during normal operation of the electrolyzer. In other words, during normal operation, pressurized electrolyte water may be pumped into the accumulator **138** but may not leave the accumulator **138**, until the outlet valve is opened. As will be described in more detail below, using a normally-open outlet valve enables the accumulator **138** to be used as a safety measure, e.g. during power cuts.

[0072] The first gas collection chamber **106** comprises a first drain port **142**. The second gas collection chamber **110** comprises a second drain port **144**. The first drain port **142** is connected to the pump **134** of the electrolyte water supply circuit **130** via a first drain line **146**. The second drain port **144** is connected to the pump **134** via a second drain line **148**. Electrolyte water permeating the first electrode **108** during operation of the electrolyzer **100** may be drained from the first gas collection chamber **106** via the first drain port **142** and the first drain line **146** respectively. Similarly, electrolyte water permeating the second electrode **112** during operation of the electrolyzer **100** may be drained from the second gas collection chamber **110** via the second drain port **144** and the second drain line **148** respectively. In multicell arrangements any drain port can act for one or more of the cells rather than being required by each cell.

[0073] In the embodiment of FIG. 1, the first and second drain ports **142, 144** are arranged typically at a bottom end of the first and second gas collection chambers **106, 110**. In other words, the drain ports **142, 144** are arranged below the first and second gas outlet ports **122, 124** of the first and second gas collection chambers **106, 110** and below the first and second electrodes **108, 112**. In other words, both the first and the second gas collection chambers **106, 110** comprise drain wells **141, 143** arranged typically at a bottom end of the gas collection chambers **106, 110**. As can be seen from FIG. 1, the housing **102** is thus substantially H-shaped but does not need to be, it can be multicell or arranged similar to a plate heat exchanger design. The device can be made of multiple electrodes usually arranged in parallel to each other to form a stack of electrodes.

[0075] During operation of the electrolyzer **100**, electrolyte water that has permeated the first or second electrode **108, 112** will collect within the drain well **141, 143** at the bottom of the respective gas collection chambers **106, 110**, whereas the gases produced during the electrolysis, will rise and pressurize the gas collection chambers **106, 110**.

[0076] As long as the drain ports **142, 144** are covered with electrolyte water, the drain ports may be opened for drainage of the electrolyte water from the gas collection chambers **106, 110** without inadvertently removing gases via the drain ports **142, 144**. In some embodiments, the electrolyzer **100** may comprise electrolyte water level sensors arranged within the gas collection chambers **106, 110** to determine if sufficient levels of electrolyte water are available within the gas chambers **106, 110** in order to safely open one or both of the drain ports **142, 144**. These levels may or may not be actively controlled with active movement of electrolyte water to the collection chambers from areas of the electrolyte circulation system with capacity including **138** or as active top up from external water source. The levels may or may not be controlled using active pressure variation and control within the system.

[0077] The first and second gas collection chambers **106, 110** both comprise gas pressure gauges **150, 152**. A first gas pressure gauge **150** is configured to determine the gas pressure within the first gas collection chamber **106**. A second gas pressure gauge **152** is configured to determine a second gas pressure within the second gas collection chamber **110**. In some embodiments, the first and second gas gauges may be incorporated into a single device.

[0078] The electrolyzer **100** comprises a first electrical terminal **154** and a second electrical terminal **156**. In the example of FIG. 1, the first terminal **154** is a negative terminal, whereas the second terminal **156** is a positive terminal. However, it will be appreciated that the polarity of the first and second terminals may be swapped, such that the first terminal **154** is a positive terminal and the second terminal **156** is negative. This is because, in some embodiments, the electrolyzer is symmetrical on either side of the electrolyte chamber **104**.

[0079] The terminals **154, 156** are connected to the housing **102** of the electrolyzer **100**. In particular, the first terminal **154** is connected to the first gas collection chamber **106**. The second terminal is connected to the second gas collection chamber **110**. Accordingly, the first terminal **154** is electrically connected to the first electrode **108**, which is electrically connected to the first gas collection chamber **106** via its second surface **116**. The second terminal **156** is connected to the second gas collection chamber **110**, which in turn is connected to the second electrode **112** via the second surface **120** of the second electrode **112**. Of course, it will be appreciated that the terminals **154, 156** may also be connected to the electrodes **108, 112** directly, rather than via the housing **102** of the electrolyzer **100**.

[0080] The terminals **154, 156** are connectable to a power source, e.g. a direct current power source, to apply a current across the electrodes **108, 112**. If the terminals **154, 156** are connected to the power source, current will flow between the first and second electrode **108, 112** via the electrolyte water within the electrolyte chamber **104**, thereby activating the electrolysis process for decomposition and separation of electrolyte water into oxygen and hydrogen, as will be explained in more detail below.

[0081] As mentioned above, in the embodiment of FIG. 1, the first terminal 154 is a negative terminal and the second terminal 156 is a positive terminal. Accordingly, the first electrode 108 of the embodiment in FIG. 1 is the cathode of the electrolyzer 100, whereas the second electrode 112 is the anode of the electrolyzer 100. The first electrode 108 is permeable to gas, particularly hydrogen. The second electrode 112 is permeable to gas, particularly oxygen.

[0082] In one example, the first and second electrodes 108, 112 comprise different porosities. In some embodiments, the porosity of the first electrode 108 may be about half of the second electrode 112. The first electrode 108 may have a porosity below 0.3 μm . The second electrode 112 may have a porosity below 0.6 μm .

[0083] In the above example, the first gas collection chamber 106 is configured to receive hydrogen gas, whereas the second gas collection chamber 110 is configured to receive oxygen gas.

[0084] In some embodiments, the anode, i.e. the second electrode 112 in FIG. 1, may be magnetic. Magnetism of the anode may improve formation of oxygen in and on the second electrode 112. To this end, the second electrode 112 may be formed as a permanent mag net. Alternatively, the second electrode may be formed as an electric magnet, activated when an electric current is applied across the two electrodes 108, 112 during operation of the electrolyzer 100.

[0085] The electrolyzer 100 further comprises a control unit 160, schematically represented in FIG. 1. The control unit may be connected to the first and/or second gas pressure gauges 150, 152 to receive gas-pressure-data representative of a gas pressure within the first and/or second gas collection chamber 106, 110.

[0086] The control unit 160 may be connected to the first and second drain valves 142, 144 for controlling operation of the first and second drain valves 142, 144. The control unit may be connected to the first and second gas outlet ports 122, 124 for controlling the operation of the first and second gas outlet ports 122, 124. The control unit 160 may be connected to the power source (not shown) for controlling the supply of electrical power to the first and second electrodes 108, 112. The control unit 160 may be connected to an outlet valve of the accumulator 138. The control unit 160 may be connected to the pump 134 and the electrolyte water pressure gauge 136.

[0087] The control unit may be connected to any of the above devices via control wires or wirelessly as is well known in the art. The control unit may either be locally arranged together with the housing 102 of the electrolyzer or remotely, e.g. in a centralized control office.

[0088] The control unit 160 is configured to control a pressure drop across at least one of the permeable electrodes 108, 112. In one embodiment, the control unit 160 is configured to control the electrolyte pressure in the electrolyte chamber 104 relative to a gas pressure in the first or second gas collection chamber 106, 110. The control unit 160 may be configured to control the electrolyte water pressure in the electrolyte chamber 104 to be higher than a gas pressure in the first or second gas collection chambers 106, 110. In other words, the control unit 160 is configured to maintain a pressure drop between the electrolyte chamber 104 and the gas collection chambers 106, 110. In some examples, the control unit 160 is configured to control the electrolyte water pressure in the electrolyte chamber 104 to be at least 5 bar higher than a gas pressure in the first and/or

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[0090] Maintaining a pressure drop of at least 5 bar between the electrolyte chamber 104 and the first and/or second gas collection chamber 106, 110 causes electrolyte water to permeate the first electrode 108 and/or the second electrode 112 and thus to flow between the electrolyte chamber 104 and the first and/or second gas collection chamber 106, 110 together with the hydrogen gas produced at the first electrode.

[0091] Causing the above electrolyte water flow across the first and/or second electrodes 108, 112 significantly increases the efficiency of the electrolyzer 100.

[0092] In order to maintain the required pressure drop across one or both of the electrodes 108, 112, the control unit 160 of FIG. 1 is configured to receive gas-pressure-data representative of a gas pressure within the first or second gas collection chambers 106, 110 respectively. In the example of

FIG. 1, the gas-pressure-data may be pressure readings supplied by the first and/or second gas pressure gauges **150, 152**.

[0093] On the basis of the gas-pressure-data, the control unit will determine a desired electrolyte water pressure within the electrolyte chamber **104**. In some examples, the control unit may add a preselected amount of pressure to the gas pressure indicated by the gas-pressure-data in order to determine the desired electrolyte water pressure. In some embodiments, the control unit **160** may determine a desired electrolyte water pressure that is at least 5 bar higher than the gas pressure within the first gas collection chamber **106**.

[0094] The control unit may then control the electrolyte water supply circuit **130** to supply electrolyte water to the electrolyte chamber **104** until the desired electrolyte water pressure is reached. In the example of FIG. 1, the control unit may be configured to activate the pump **134** to supply electrolyte water to the control chamber **104** until the desired electrolyte water pressure has been reached. To this end, the control unit may receive electrolyte-pressure-data representative of a pressure of electrolyte water within the electrolyte chamber **104**. In the embodiment of FIG. 1, the electrolyte-pressure-data comprises pressure readings supplied by the pressure gauge **136** arranged downstream of the pump **134**.

[0095] The control unit may control activation of the pump **134** via a control loop, based on electrolyte-pressure-data provided by the pressure gauge **136**. For example, the control unit may control the electrolyte water pressure within the

[0096] text missing or illegible when filed control chamber **104**, represented by the pressure readings of the pressure gauge **136**, via a PID control loop.

[0097] It should be appreciated that a pressure drop between the electrolyte chamber **104** and the gas collection chambers **106, 110** will vary continuously as the electrolyzer **100** is operated. This is because, during operation of the electrolyzer, i.e. when pressurized electrolyte water is available in the electrolyte chamber **104** and a current is applied across the two electrodes **108, 112**, hydrogen and oxygen gases are produced and added to the first and second gas collection chambers **106, 110** continuously. Accordingly, in this example, as long as the first and second gas outlet ports **122, 124** remain closed, the gas pressure within the gas collection chambers **106, 110** will continue to rise. This exemplary rise in gas pressure will be determined by the control unit **160** on the basis of the gas-pressure-data. The control unit **160** will then determine a new, higher, desired electrolyte water pressure and control the pump **134** to increase the electrolyte water pressure within the electrolyte chamber **104** and match said increased desired electrolyte water pressure. The control unit **160** may continuously adjust the pressure within the electrolyte chamber **104** as long as the gas pressure within the first or second gas collection chamber **106, 110** rises.

[0098] The control unit **160** may also control the gas pressure within the first and/or second gas collection chamber **106, 110**. In the example of FIG. 1, the control unit may be configured to control an operation of the first and/or second gas outlet port **122, 124**. The control unit **160** may open and close the first gas outlet port **122** to control the (hydrogen) gas pressure within the first gas collection chamber **106**. The control unit **160** may open and close the second gas outlet port **124** to control the (oxygen) gas pressure within the second gas collection chamber **110**. In some embodiments, the control unit **160** may receive a first desired gas pressure for the first gas collection chamber **106** and a second desired gas pressure for the second gas collection chamber **110**. The first and second desired gas pressures may be selected by an operator. In some examples, the desired gas pressures may be determined directly by apparatus using the hydrogen and oxygen gases provided by the electrolyzer **100**.

[0099] In another embodiment, one or both of the gas outlet ports may comprise pressure relief valves configured to open automatically once the gas pressure in the first or second gas collection chamber **106, 110** exceeds a set pressure. In this example, the pressure within the gas collection chambers will be determined by the set pressure of the pressure relief valves of the first and second gas outlet ports **122, 124** respectively. In some embodiments, the set pressure of the pressure relief

valves may be adjustable, e.g. via the control unit **160**.

[0100] In both embodiments described above, the control unit **160** may be configured to maintain a gas pressure in the first and second gas collection chamber **106, 110** at 100 bar to 1000 bar. If the gas pressure within the gas collection chambers **106, 110** is maintained at 100 bar, the control unit may set a desired electrolyte water pressure of 105 bar or more to allow for some electrolyte water to pass through the first and/or second electrode **108, 112** as has been described above.

[0101] It should be noted that electrolyte water passing through the first and second electrodes **108, 112**, due to the pressure drop between the electrolyte chamber **104** and the gas collection chambers **106, 110**, may be drained back into the electrolyte water supply circuit **130**, e.g. intermittently, via the above drain ports **142, 144**. The control unit **160** may be configured to control such electrolyte water drainage operation. To this end, the control unit **160** may be connected to water level sensors (not shown) arranged within the first and/or second gas collection chambers **106, 110**. The water level sensors may provide the control unit with water-level-data representative of the water levels within the first and second gas collection chambers **106, 110** particularly within the drain wells **141, 143**. If, on the basis of the water-level-data, the control unit determines that the electrolyte water levels are sufficiently high to fully cover the first and/or second drain port **142, 144**, the control unit **160** may temporarily open one or both of the drain ports **142, 144** to drain electrolyte water from the gas collection chambers **106, 110**. In one embodiment, the control unit may be configured to open the drain ports **142, 144** for a predetermined amount of time, e.g. several seconds, if the water level has reached a water-level-threshold. In other embodiments, the control unit will drain electrolyte water until the electrolyte water levels have fallen below a predetermined water-level-threshold.

[0102] The control unit may also be configured to control a power source (not shown) attached to the electrodes **108, 112** via the terminals **154, 156**. The operator or controller can alter the amperage according to the type of electrolysis and the type of electrode used and other variables such as cell gap. The control unit may be configured to set the voltage provided by the power source to be set at the desired voltage. The desired voltage may be designated by the operator. The amperage and voltage can be fixed, manually set or variably controlled by the control unit **160**.

[0103] In some embodiments, the control unit may be configured to supply electrical power to the electrodes **108, 112** only once the desired pressure drop across the electrodes **108, 112** has been achieved. In other words, the control unit **160** may monitor a pressure difference between one of the gas collection chambers **106, 110** and the electrolyte chamber **104**. Once the pressure difference exceeds a selectable first pressure-threshold, the control unit may activate the power supply to apply a DC current across the electrodes **108, 112** and the electrolyte water within the electrolyte chamber **104** to start operation of the electrolyzer **100**. The control unit **160** may be configured to de-activate the power supply whenever the pressure difference falls below a second pressure-threshold. The second pressure threshold may be the same as or lower than the first pressure-threshold.

[0104] As mentioned above, the accumulator **138** may be arranged such that, during normal operation, electrolyte water may only be added to the accumulator. In other words, pressurized electrolyte water may not leave the accumulator **138** during normal operation. The accumulator may comprise a normally-open outlet valve. The control unit **160** may be configured to maintain the normally-open outlet valve in a closed position during normal operation of the electrolyzer. In case of a power cut, i.e. when the pump **136** and potentially the control unit **160** are inactive, the normally-open outlet valve of the accumulator will open such that the accumulator is connected to the electrolyte chamber **104**. Accordingly, the electrolyte water stored in the accumulator **138** will be able to flow into the electrolyte chamber to maintain the pressure drop of across the first and second electrodes **108, 112** for a period of time, despite the power cut. In this scenario, no current is applied across the electrodes **108, 112** such that the electrolysis process will come to a stop and no further gases will be produced. Due to the elevated pressure maintained within the electrolyte

chamber **104** by means of the accumulator **138**, however, electrolyte water will be pushed through the two permeable electrodes **108**, **112** despite the power cut. As a consequence, remaining gas bubbles formed during operation will be flushed out of the pores of the electrodes via the prolonged flow of electrolyte water after the operation of the electrolyzer **100** has come to a stop.

[0105] The control unit **160** may be configured to perform a forward flushing operation for removing debris or bubbles that are stuck in the pores of the electrodes **108**, **112**. During the forward flushing operation, the control unit **160** may interrupt any current applied across the first and second electrodes **108**, **112** and maintain the desired electrolyte water pressure within the electrolyte chamber **104**. Due to the lack of electric current applied across the

[0106] text missing or illegible when filed electrodes, no more hydrogen or oxygen bubbles will be produced at the electrodes **108**, **112**. Electrolyte water that is still being pumped through the electrodes **108**, **112** will remove debris and remaining bubbles as it passes through the electrode's pores. During the forward flushing operation, the control unit may hold open the drain ports **142**, **144**.

[0107] In another embodiment, the control unit **160** may be configured to perform a backward flushing operation for removing debris or bubbles that are stuck in the pores of the electrodes **108**, **112**. During the backward flushing operation, the control unit may pump electrolyte water into the first and/or second gas collection chambers **106**, **110** via their respective drain ports **142**, **144**. To this end, the pump **134** may be a bi-directional pump. Electrolyte water pumped into the gas collection chambers **106**, **110** via the drain lines **146**, **148** and the drain ports **142**, **144** will then permeate the electrodes **108**, **112** from their second surfaces **116**, **120** towards their first surfaces **114**, **118** and into the electrolyte chamber **104**. Contaminated flushing fluid (e.g. electrolyte water) may be removed from the electrolyte water supply circuit **130** via the vent line **105**, during the reverse flushing operation. Flushed fluid may also be removed at or before the pump **134**, i.e. between port **126** and pump **134**. During the flushing operation, the control unit **160** may be configured to close the inlet port **126** and open an outlet port of the electrolyte chamber **104** that is connected to the vent line **105**. Electrolyte water removed from the electrolyte supply circuit **130** may be replenished via the electrolyte water supply line **146**.

[0108] The control unit may be configured to perform the forward or backward flushing operations automatically, i.e. at predetermined time intervals, or in response to an operator's command.

[0109] Turning to FIG. 2, there is shown a schematic flow chart of an embodiment of a method **200** for controlling an electrolyzer. The electrolyzer may comprise two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water.

[0110] In a first step **202**, the method comprises supplying electrolyte water to the at least one permeable electrode. In a second step **204**, the method comprises controlling a pressure drop between opposite sides of the at least one permeable electrode. To this end, the method may either comprise increasing the electrolyte water pressure upstream of the permeable electrode or decreasing a gas pressure, e.g. hydrogen pressure, downstream of the at

[0111] The pressure drop may be 5 bar or more.

[0112] Preferences and options for a given aspect, feature or parameter of the disclosure should, unless the context indicates otherwise, be regarded as having been disclosed in combination with any and all preferences and options for all other aspects, features and parameters of the disclosure.

[0113] It will be appreciated that although, in the embodiments described above, a single control unit was mentioned for the control of various aspects of the electrolyzer of the present disclosure, the electrolyzer may also include a plurality of control units controlling some of the aforementioned aspects and, preferably, communicating with each other.

Claims

1. An electrolyzer for generating hydrogen, the electrolyzer comprising: a housing comprising an electrolyte chamber; two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, wherein the at least one permeable electrode has a first surface facing the electrolyte chamber and a second surface facing a first gas collection chamber; an electrolyte supply circuit for supplying electrolyte water to the electrolyte chamber; and a control unit for controlling a pressure drop across the at least one permeable electrode, between the electrolyte chamber and the first gas collection chamber.
2. The electrolyzer of claim 1, wherein the control unit is configured to: receive gas-pressure-data representative of a gas pressure within the first collection chamber; determine, on the basis of the gas-pressure-data, a desired electrolyte water pressure within the electrolyte chamber, control the electrolyte supply circuit to set the pressure within the electrolyte chamber to the desired electrolyte pressure.
3. The electrolyzer of claim 2, wherein the control unit is configured to set the desired electrolyte pressure to be higher than a gas pressure within the first gas collection chamber, wherein a difference between the electrolyte pressure and the gas pressure is above a selectable first pressure-threshold.
4. The electrolyzer of claim 3, wherein the pressure threshold is at least 5 bar.
5. The electrolyzer of claim 3, wherein the control unit is configured to apply a current across the electrodes if the pressure difference between the first gas collection chamber and the electrolyte chamber is above the selectable pressure-threshold.
6. The electrolyzer of claim 1, wherein the electrolyzer comprises a pressure control valve for regulating a gas pressure within the first gas collection chamber.
7. The electrolyzer of claim 2, wherein the electrolyte supply circuit comprises a hydraulic pump for supply of electrolyte water to the electrolyte chamber, and wherein the control unit is configured to: receive electrolyte-pressure-data representative of a pressure of electrolyte water within the electrolyte chamber; and control, on the basis of the electrolyte-pressure-data, an operation of the pump.
8. The electrolyzer of claim 1, wherein the at least one permeable electrode has a porosity of 0.3 μm or less, preferably 0.2 μm or less.
9. The electrolyzer of claim 8, wherein both electrodes are permeable to gases produced by the decomposition of electrolyte water, and wherein the electrodes have different porosities.
10. The electrolyzer of claim 1, wherein the at least one permeable electrode is made of steel, preferably sintered steel.
11. (canceled)
12. The electrolyzer of claim 1, wherein at least one of the electrodes, preferably the anode, is magnetic.
13. The electrolyzer of claim 1, wherein the electrolyte supply circuit comprises a pressure storage reservoir selectively connectable to the electrolyte chamber, wherein the reservoir is preferably connected to the electrolyte chamber via a normally-open outlet valve, which is closed during normal operation of the electrolyzer.
14. (canceled)
15. The electrolyzer of claim 1, comprising a control unit configured to: receive a forward-flushing-command; interrupt any current applied across the electrodes and increase a pressure in the electrolyte chamber to be more than 5 bar above a gas pressure within the first gas chamber in response to the forward-flushing command.
16. The electrolyzer of claim 1, wherein the electrolyte chamber comprises an electrolyte gap located between the two electrodes, wherein the gap preferably has a width of 0.1 mm to 10 mm.
17. The electrolyzer of claim 1, wherein the first gas collection chamber comprises a first drain port

for draining electrolyte water that has passed through the at least one permeable electrode back into the electrolyte supply circuit.

18. The electrolyzer of claim 17, wherein the control unit is configured to: receive a backward-flushing-command; interrupt any current applied across the electrodes and supply electrolyte water to the first gas collection chamber via the first drain port to reverse flush the at least one permeable electrode from the first gas collection chamber towards the electrolyte chamber.

19. A method of controlling an electrolyzer for generating hydrogen from water, the electrolyzer comprising two electrodes for decomposition of electrolyte water, at least one of the electrodes being permeable to gases produced by the decomposition of electrolyte water, the method comprising: supplying electrolyte water to the at least one electrode; controlling a pressure drop between opposite sides of the at least one electrode.

20. The method of claim 19, further comprising maintaining a pressure drop between opposite sides of the at least one electrode that is higher than a selectable pressure-threshold.

21. The method of claim 20, further comprising: Applying electricity across the electrodes if the pressure drop across opposite sides of the at least one electrode is higher than the selectable pressure-threshold.

22. The method of claim 21, wherein the selectable pressure threshold is 5 bar or more.
